Thapar Institute of Engineering & Technology

Department of Computer Science and Engineering

Algo Recall

Software Engineering Project Report

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Chapter 1

Project Selection Phase

1.1 Software Bid

Project Description

Algo Recall is a web-based application designed to help students manage and revise Data Structures and Algorithms (DSA) questions they have already solved. The system uses confidence-based revision scheduling to recommend questions for revision and helps learners track their progress and consistency over time.

Proposed Solution

The proposed system provides a centralised platform where learners can store solved DSA questions, update their confidence levels after revision, and access a personalised dashboard. It maintains revision history and provides topic-wise analytics and consistency statistics to help learners identify areas that require more practice.

Expected Deliverables

- User registration and login.
- Question management, including adding, updating, deleting, searching, and filtering questions.
- Confidence-based revision scheduling and daily revision recommendations.
- Revision history and progress tracking.
- Topic-wise analytics and consistency statistics.

Project Scope

The first version of Algo Recall focuses on question management, revision scheduling, and progress monitoring. Features such as online code execution, contest management,

AI-generated solutions, and automatic LeetCode synchronisation are outside the scope of this version.

Expected Outcome

The project aims to provide learners with a structured and manageable revision process, encourage consistent practice, and make it easier to identify DSA topics that need further revision.

1.2 Project Overview

Algo Recall helps learners organise questions they have already solved and decide which questions to revise next. Learners can record confidence levels, view suggested revisions, and track revision progress and consistency.

The V1 scope includes registration and login, question management, revision, revision history, search and filtering, and analytics. Online judge or code execution, contest management, AI-generated solutions, and automatic LeetCode synchronisation are outside the stated V1 scope.

Chapter 2

Analysis Phase

2.1 Use Cases

2.1.1 Use-Case Diagrams

The diagram contains one actor, *User (Learner)*, and twelve use cases inside a system boundary labelled *Algo Recall*. Developers are not included as actors because development and maintenance are not normal runtime interactions.

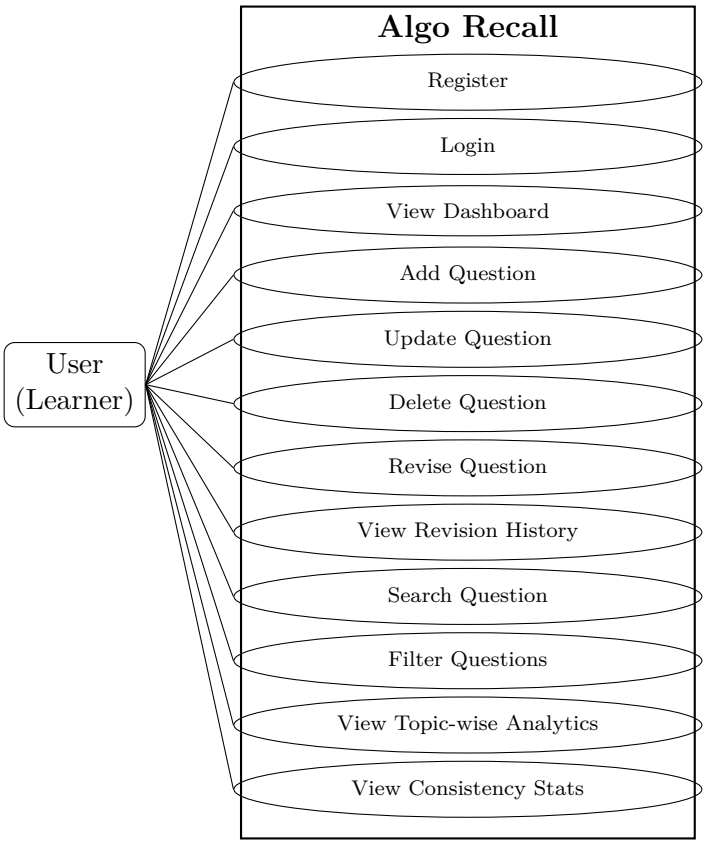


Figure 2.1: Algo Recall V1 use-case diagram

Use cases and descriptions

No.	Use Case	Description
1	Register	Creates a new account on Algo Recall.
2	Login	Authenticates the user and grants access to their account.
3	View Dashboard	Displays the day's suggested revision questions and revision completion progress.
4	Add Question	Stores a solved DSA question with its details, confidence level, and revision information.
5	Update Question	Modifies the stored details of an existing question.
6	Delete Question	Removes a question from the user's stored collection.
7	Revise Question	Marks a suggested question as revised and updates its confidence level and revision details.
8	View Revision History	Displays previous revision sessions and their recorded details.
9	Search Question	Finds a stored question using its name or other searchable information.
10	Filter Questions	Filters stored questions by topic, difficulty, or other supported attributes.
11	View Topic-wise Analytics	Displays revision and confidence statistics for different DSA topics.
12	View Consistency Stats	Shows revision activity over time so the user can track consistency.

2.1.2 Use Case Templates

Use Case ID: UC-07
Name: Revise Question
Primary actor: User (Learner)

Preconditions

- The learner is logged in.
- The question exists in the learner's collection.

Trigger: The learner chooses a suggested question to revise.

Main flow

1. The learner opens the dashboard and selects a suggested question.
2. The learner revises the question.
3. The learner submits a confidence rating.
4. The system updates the question's confidence and revision details.
5. The system records the revision in history and updates the next-revision schedule.

Alternative flow: If the confidence rating is invalid or the question cannot be found, the system returns an error instead of recording the revision.

Postconditions: The question and revision history reflect the recorded revision. The exact scheduling formula is not specified in the V1 reference.

2.2 Activity Diagram and Swimlane Diagrams

This activity diagram illustrates the high-level revision workflow. It can be expanded into swimlanes for the learner, application, and database if required.

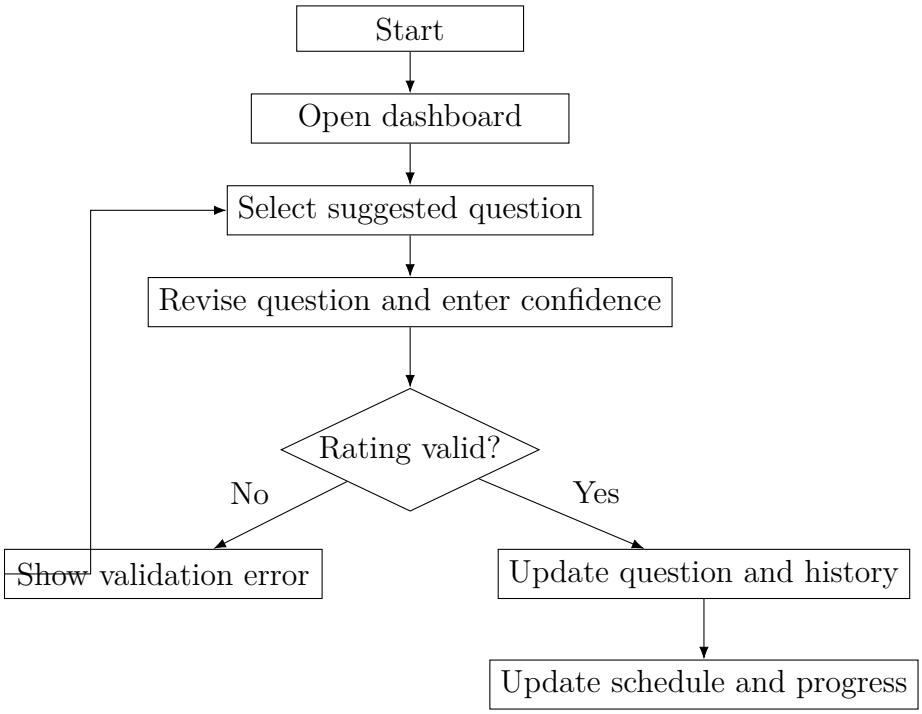


Figure 2.2: Activity diagram for revising a question

2.3 Data Flow Diagrams (DFDs)

These diagrams are illustrative and show broad data interactions. They should be checked against the final implementation before submission.

2.3.1 DFD Level 0

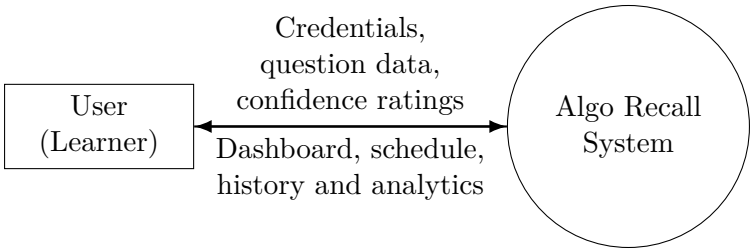


Figure 2.3: DFD Level 0: Algo Recall context diagram

2.3.2 DFD Level 1

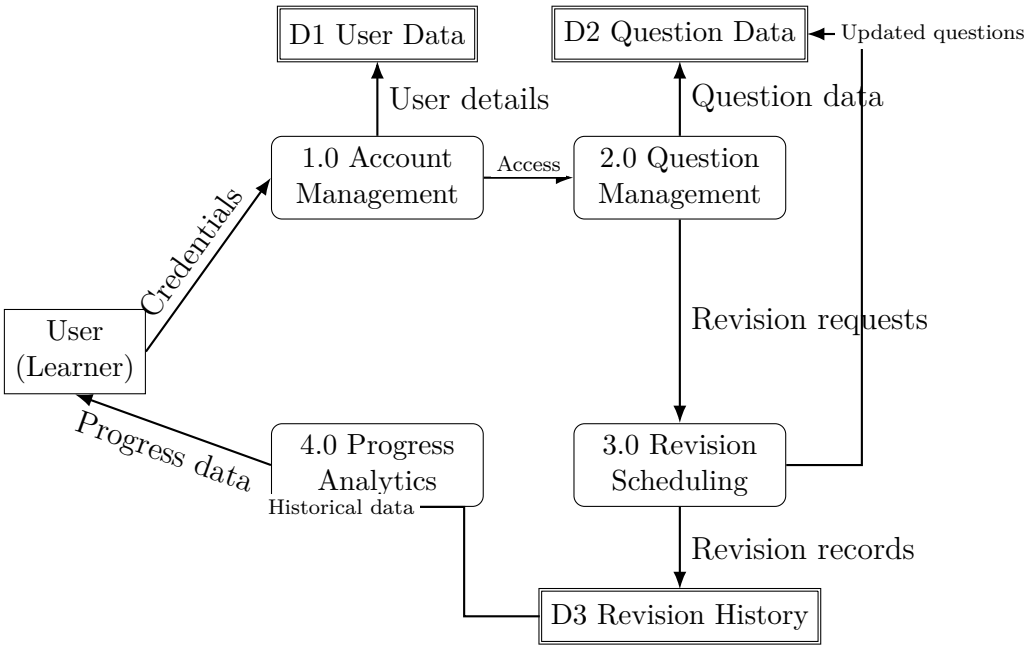


Figure 2.4: DFD Level 1: Main processes and data stores

2.3.3 DFD Level 2

This example expands the revision-scheduling process:

1. Receive the selected question and confidence rating.

2. Validate the rating and verify question ownership.
3. Calculate the next revision date and priority.
4. Save the updated question and create a revision-history record.
5. Return the updated question and confirmation to the learner.

The V1 reference states that the dashboard should suggest approximately 5–7 questions per day. Unfinished suggested questions should carry forward alongside newly scheduled questions, and the visible workload should not grow without control. The exact mechanism for controlling this workload is not specified and should be documented once decided.

2.4 Software Requirement Specification in IEEE Format

This concise draft is based on the supplied V1 scope. Expand it if the course requires a specific IEEE SRS template.

1. Introduction

Purpose: Specify the user-facing requirements for Algo Recall V1.

Scope: Algo Recall allows learners to manage solved DSA questions, revise them using confidence-based scheduling, and monitor revision progress and consistency.

2. Overall Description

Primary user: User (Learner).

Included functions: Registration, login, dashboard, question management, revision, revision history, search, filtering, topic-wise analytics, and consistency statistics.

Excluded from V1: Online judge or code execution, contest management, AI-generated solutions, and automatic LeetCode synchronisation.

3. Functional Requirements

- FR-1: The system shall allow a learner to register for an account.
- FR-2: The system shall allow a learner to log in.

- FR-3: The system shall display suggested revision questions and completion progress.
- FR-4: The system shall allow a learner to add, update, and delete stored questions.
- FR-5: The system shall allow a learner to record a revision and confidence rating.
- FR-6: The system shall provide revision history.
- FR-7: The system shall support question search and filtering by supported attributes.
- FR-8: The system shall display topic-wise revision and confidence statistics.
- FR-9: The system shall display revision activity over time.
- FR-10: The dashboard should target approximately 5–7 suggested questions per day, carry unfinished questions forward, and prevent the visible workload from growing without control.

4. Non-Functional Requirements

- Usability: The system should provide a simple and user-friendly interface for managing DSA questions and revision activities.
- Security: Users should only be able to access and modify their own data.
- Reliability: The system should store question details and revision history accurately.
- Performance: The system should handle routine operations such as adding, updating, searching, and revising questions efficiently.
- Maintainability: The system should have a modular code structure that allows future updates and modifications easily.

2.5 User Stories and Story Cards

The following sample stories are based on the listed V1 use cases.

ID	User Story	Priority
US-01	As a learner, I want to register and log in so that I can access my account.	High

US-02	As a learner, I want to add, update, and delete DSA questions so that I can manage my solved-question collection.	High
US-03	As a learner, I want to record my confidence after revising a question so that its revision details can be updated.	High
US-04	As a learner, I want to see today's suggested questions and completion progress so that I can plan my revision.	High
US-05	As a learner, I want to search and filter questions so that I can find a question quickly.	Medium
US-06	As a learner, I want to view revision history so that I can review previous sessions.	Medium
US-07	As a learner, I want topic-wise analytics and consistency statistics so that I can monitor progress.	Medium

Example story card — US-03: Record confidence after revision

Acceptance criteria

- The learner can submit a confidence rating.
- The system validates the rating according to the application's supported range.
- A valid rating updates the question's confidence and revision details.
- The system records the revision in revision history.

Bibliography

[1] OpenJS Foundation, *Node.js Documentation*. Available at: `https://nodejs.org/docs/`

[2] Express.js, *Express Documentation*. Available at: `https://expressjs.com/`

[3] MongoDB, *MongoDB Manual*. Available at: `https://www.mongodb.com/docs/`